Qinghua Sun

21-402, No. 268 West Wangjiang Rd, Shushan District, Hefei, 230022, Anhui, China 15105510419@163.com

November 27th, 2002 +86 15105510419

EDUCATION HISTORY

University of Electronic Science and Technology of China Sep 2021 – Jun 2025

- Glasgow College; Communication Engineering Joint Program with the **University of Glasgow**
- Cumulative GPA: 3.56/4.0; Double B.Eng. Degrees Granted June 2025

PLAN OF GRADUATE STUDY

University of California San Diego Mar 2026 – Dec 2027

- Program: MS Electrical and Computer Engineering
- Major：Signal and Image processing

RESEARCH EXPERIENCE

Team Leader – Design of a STM32 Smart Line-Patrolling Vehicle Apr – Jun 2024

- Developed a visual line-tracking algorithm using image processing; divided 6 ROIs to extract track edges, fitted the track centerline, and calculated heading deviation for stable, centered steering. Integrated HSV-based color segmentation to detect traffic lights (red stop/green go) and arrow indicators for three-way-intersection path selection, achieving >91% recognition accuracy.
- •Implemented multi-scenario obstacle and pedestrian detection: used ultrasonic sensing (±2 cm accuracy) to trigger avoidance when obstacles were within 30 cm, enabling smooth detour maneuvers. Added zebra-crosswalk logic to detect pedestrian intrusion and execute automatic stopping with <200 ms response.
- •Programmed vehicle control on STM32 using PID closed-loop speed regulation to reduce tracking error to ±3 cm and improve turning stability; dynamically loaded preset speed profiles based on arrow commands to ensure accurate lane entry.
- Served as the team leader of six, managed the progress, arranged meetings, and delivered presentations for defense

Creator - Deep Learning Framework for Battery Abnormality Detection Jan - Jun 2025

- Led the design and development of Dynamic Anomaly Detector (DAD), an end-to-end deep learning framework for EV battery fault detection. Integrated graph attention and incremental learning to overcome limitations in open-set fault identification and aging adaptability.
- Built a 1D-CNN autoencoder to model normal battery behavior using the nonlinear mapping between inputs (SOC, current) and outputs (voltage, temperature), enabling unsupervised anomaly detection trained solely on healthy operating data.
- Proposed a dynamic percentile thresholding strategy (95th–99th) to adaptively interpret reconstruction errors under environmental changes and aging-induced distribution shifts, reducing false alarms by up to 40% compared with fixed thresholds.
- Constructed a comprehensive evaluation framework using a multi-source dataset of 347 EVs and 90,000+ charging cycles, applying five-fold cross-validation. DAD surpassed baselines (e.g., LSTM-AD) in AUC, recall, and F1, achieving an average AUC gain of 3.2 points and significantly improving generalization to unseen faults.

INTERNSHIP EXPERIENCE

Product & Testing Intern - Shanghai Yunwu Intelligent Technology Co., Ltd.(feeling AI) Oct 2025 – Mar 2026

- Spearheaded end-to-end definition and iteration of Flimo's generative AI features from 0 to 1, with a focus on "Create" (AI-driven story scripting and directorial control) and "Space" (real-time conversational interaction with emotionally expressive 3D avatars). Translated user insights and competitive analysis into actionable product requirements—such as "lowering the barrier to content creation" and "enhancing character personality consistency"—and collaborated closely with LLM, UE engine, and algorithm teams to operationalize these goals into technical solutions, including prompt engineering, fine-tuning strategies, and emotion-to-animation pipelines.
- Led end-to-end evaluation of AI-powered experiences by designing test cases across dimensions including text generation quality, response coherence, and multimodal interaction fluency. Identified and drove resolution of 50+ UX-critical issues (e.g., mismatched avatar gestures vs. dialogue semantics, narrative logic gaps), coordinating cross-functional alignment sessions with Algorithm, Backend, Art, Infra, and iOS teams to define root causes and validate fixes against product vision and user expectations.
- Established a scalable AI product development rhythm: initiated weekly tripartite syncs among Product, Algorithms, and Engineering to align on AI-specific OKRs (e.g., prompt optimization, SFT data labeling, inference latency reduction). Tracked daily progress via structured logs, proactively flagged resourcing bottlenecks, and ensured on-time, high-quality delivery of AI features across release cycles.
- Documented institutional knowledge to bridge the AI literacy gap: authored internal guides including "AI Feature Acceptance Criteria" and "Taxonomy of Multimodal Interaction Issues," enabling non-AI stakeholders to understand model capabilities and limitations.

HONORS & SKILLS & TESTS

- University Summer 'Lighting' Volunteer Team Future Merit Scholarship
- 2nd Class Prize Hainan Province College Student Entrepreneurship and Innovation Competition (30000 RMB)
- University of Electronic Science and Technology of China Excellent Student Cadre (UESTC)
- Programming Languages: Proficient in C, Python, MATLAB, HTML, CSS, Javascript
- Software Tools: Experienced with MATLAB, STM32CubeIDE
- Language and Tests: Native Chinese; IELTS Overall 7.5; GRE 321 (Quantitative 170 + Verbal 151 + Writing 3.5)